**4** A non-uniform L-shaped beam of weight 10 N is attached to a wall using a hinge. The longer part of the beam is 50 cm and the shorter part of the beam is 10 cm. The weight of the beam acts at a point on the beam that is at a horizonal distance of *x* away from the hinge. The beam is held at rest using a cord, such that the longer part of the beam is horizontal. *AO* is horizontal while *OC* is vertical. The tension in the cord is 5.0 N, at an angle of 60° from *OC*, as shown in Fig. 4.1.

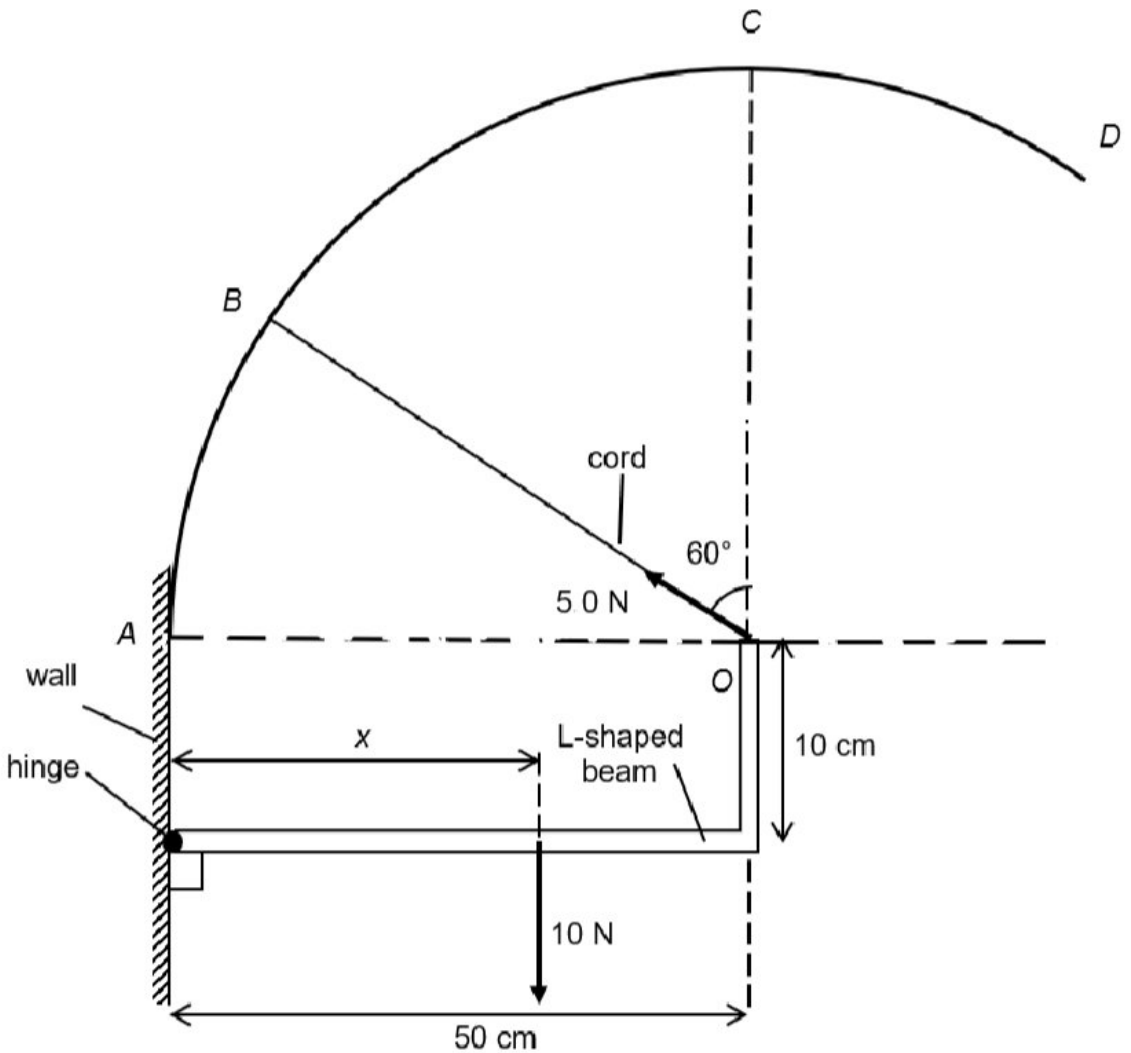



**Fig. 4.1**

**(a)** On Fig. 4.1, draw and label the contact force *R* acting on the L-shaped beam due to the hinge.

[2]

**(b)** By considering the forces acting on the beam, calculate the magnitude of $R$.

magnitude of $R$ = ..... N [3]

**(c)** Determine the value of $x$.

$x$ = ..... cm [2]

**(d)** $ABCD$ are different points on a circle where the centre is at point $O$. $C$ is at the point vertically above point $O$. The cord moves along the arc of the circle such that the beam remains stationary.

**(i)** State how the magnitude of the contact force $R$ would change as the cord moves from $B$ to $C$.

.....

.....

..... [1]

**(ii)** State how the direction of the contact force $R$ would change as the cord moves from $C$ to $D$. Mathematical calculation is not required in this part.

.....

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..... [1]

[Total: 9]